Hypertension

ORIGINAL ARTICLE

Factors Associated With Discordant Blood Pressure Measures among Very Old Adults: Results From the Atherosclerosis Risk in Communities (ARIC) Study

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BACKGROUND: Home blood pressure (BP) monitoring (HBPM) is increasingly used as an alternative to office BP. However, factors influencing agreement between office and home BP among very old adults remain unclear.

METHODS: During ARIC (Atherosclerosis Risk in Communities) visit 10, participants underwent 3 automated office BP (AOBP) measurements using an Omron HEM-907XL and performed HBPM twice daily for 8 days using an Omron BP7450. Discordance was defined as a systolic BP difference of ±10 mm Hg between mean AOBP and HBPM. Multivariable regression models evaluated demographic, anthropometric, and clinical factors associated with discordance.

RESULTS: Among 792 participants (58% female; mean age, 84±3.7 years), mean systolic BP was 130.6 mm Hg (AOBP) and 129.6 mm Hg (HBPM). Despite a minimal average difference (1.0±15.7 mm Hg), 49% had ≥10 mm Hg systolic BP discordance. Higher AOBP was associated with greater discordance. Compared with females, males had lower AOBP relative to HBPM (−4.69 mm Hg [95% CI, −6.86 to −2.51]). Smaller arm circumference was associated with higher discordance (β=14.4 mm Hg [95% CI, 4.78–24.04]). Frail adults had lower AOBP relative to HBPM (β, −5.1 mm Hg [95% CI, −11.0 to 0.9]). Baseline AOBP systolic BP ≥140 mm Hg strongly predicted discordance ≥+10 mm Hg (odds ratio, 8.27 [95% CI, 5.52–12.40]). Participants aged 91 to 100 years had lower AOBP than those aged 78 to 80 years (β, −5.0 mm Hg [95% CI, −10.06 to 0.001]).

CONCLUSIONS: Among very old adults, substantial BP discordance between AOBP and HBPM was common and influenced by higher BP, age, male sex, arm circumference, and frailty. ***(Hypertension.* 2026;83:1025–1036. DOI: 10.1161/HYPERTENSIONAHA.125.26377.)** • **Supplement Material**.

Key Words: arm ■ blood pressure ■ cardiovascular disease ■ dementia

Hypertension affects over 75% of persons older than 65 years of age and represents a modifiable risk factor to prevent dementia and cardiovascular disease events among older adults.[1–5] However, current practice guidelines for hypertension screening and treatment focus heavily on blood pressure (BP) measurements obtained in clinic settings.[6,7] The focus on clinic BP is in large part based on trials demonstrating cardiovascular disease risk benefit from pharmacological management of hypertension exclusively using office BP measurements.[8] However, this reliance on clinic BP measurement represents a significant barrier for older adults,

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NOVELTY AND RELEVANCE

What Is New?

In this community-based sample of very old adults, nearly 50% of participants had a ≥10 mm Hg difference between office and home systolic blood pressure. Frailty, male sex, smaller arm circumference, and higher resting blood pressure were associated with greater discordance among very old adults.

What Is Relevant?

Both blood pressure measurement context and technique critically influence interpretation in older adults. Reliance on single-setting readings may misclassify hypertension control in clinical practice among very old adults.

Clinical/Pathophysiological Implications

Clinicians should be aware of discordance between blood pressure measures and recognize a range of factors (physiological, device, and contextual) that may contribute to discordance among very old adults.

Nonstandard Abbreviations and Acronyms

AOBP	automated office blood pressure
ARIC	Atherosclerosis Risk in Communities
BMI	body mass index
BP	blood pressure
DBP	diastolic BP
HBPM	home blood pressure monitoring
OR	odds ratio
SBP	systolic BP
SPRINT	Systolic Blood Pressure Intervention Trial

particularly those with frailty, limited transportation, or multiple medical appointments, who can face many difficulties attending clinic visits.[7,9]

Home BP monitoring (HBPM) has shown promise as an effective method to achieve timely BP treatment and control.[10–12] Furthermore, the COVID-19 pandemic catalyzed widespread adoption of HBPM programs for hypertension screening and management.[13,14] However, evidence for this practice is severely lacking for older adults. Currently, there are no HBPM cohort studies among very old populations in the United States. This is particularly concerning, as the discordance between office and home BP increases with age,[15] making older adults, who tend to be at highest risk for cardiovascular disease and dementia, especially susceptible to harms from either overly aggressive or delayed BP treatment.

In a recent ancillary study, among a community-based cohort of older adults, participants underwent automated office BP (AOBP) assessments and an HBPM protocol over an 8-day period. The objectives of the present study are to (1) characterize the association between AOBP and HBPM in a cohort of very old adults and (2) identify clinical factors related to discordance between office and HBPM. We hypothesized that resting BP, demographic characteristics (ie, age, sex, etc), physical measurements, and BP measurement procedures would be associated with discordance among older adults, with greater differences expected among frail adults and older age.

METHODS

Data Availability

Data for the present study are available from the ARIC study (Atherosclerosis Risk in Communities) but are not publicly available due to participant privacy and data use agreements. However, deidentified data may be obtained by qualified researchers on request and with approval from the ARIC Publications Committee (aricjhu@jhu.edu) and Coordinating Center (arichelp@unc.edu).

Study Design and Population

The ARIC study, established in 1987, is a long-term cohort study designed to investigate midlife cardiovascular risk factors and outcomes, initially enrolling 15 792 adults aged 45 years to 64 from 4 communities: Forsyth County, NC; Jackson, MS; Minneapolis, MN; and Washington County, MD. At the time of visit 10, all participants were at least 78 years old. Through comprehensive lab testing, physical examinations, and cardiovascular assessments, researchers collected detailed information on participant's health, demographic characteristics, and coexisting medical conditions.[16,17]

As part of ARIC visit 10, conducted from December 2022 to December 2023, an ancillary study was initiated to evaluate HBPM in older adults. Participants with seated systolic BP (SBP) <200 mm Hg and diastolic BP (DBP) <120 mm Hg measured during the clinic assessment were eligible to participate. Of the 1636 participants who attended visit 10 and completed an AOBP assessment, 1033 participated in the HBPM ancillary. We excluded 15 participants who completed in-clinic home device measurements but did not complete any at-home measurements and 56 participants who did not follow the standardized at-home protocol (see Supplemental Methods SM1 for protocol requirements). In addition, we excluded 2 participants with an arm circumference <22 or >42 cm, that is, the range of the HBPM Omron Series 10 BP7450 cuff. We also excluded 168 participants who were missing covariates of interest, resulting in a final analytic sample size of 792 participants (see Figure S1 for study flow chart). Informed consent was obtained from each participant, and the study protocol was approved by a central institutional review board at Johns Hopkins University. The Beth Israel Deaconess Medical Center institutional review board determined the present data analysis to be human subjects exempt research.

AOBP Measurement

AOBP was performed using an Omron HEM-907XL automated BP monitor, the same device used in the SPRINT (Systolic Blood Pressure Intervention Trial), with cuff sizes ranging from 17 to 50 cm.[18,19] To ensure measurement accuracy, the Omron HEM-907XL monitors were calibrated and verified semiannually using a SimCube SC-4Kit noninvasive BP simulator.[20] In addition, cuffs were regularly inspected for holes, cracks, worn-out fabric, and bladder leaks as part of quality control procedures. Participants were required to refrain from caffeinated beverages, meals, smoking, or exercise within 30 minutes before measurement. In addition, participants were also given the opportunity to void before the measurement. Each participant's arm circumference was measured per the ARIC protocol to determine the appropriate cuff size.[21] All staff involved in the study procedure were trained and certified by the investigative team.

Participants were then seated comfortably in a chair with their feet flat on the ground, back supported, and the nondominant, bare arm fully supported and positioned at heart level with the palm facing upward. This was an attended assessment, and participants were instructed to silence their phones and to remain still and silent throughout the measurement period. The Omron HEM-907XL device was configured in advance to take 3 BP readings 1-minute apart, following a 5-minute rest.[4,22]

In-Clinic Assessment Using Home Device (In-Clinic Home Device)

In the same visit (usually a few hours later), participants were provided an Omron Series 10 BP7450 device, a validated home BP monitor.[23] Before this assessment, the device was programmed to take 3 consecutive measurements with a 1-minute pause between each measurement (Truread mode). Participants were taught proper techniques for self-performed BP. The participants were then asked to rest for 5 minutes and to measure their BP according to the HBPM protocol. Study technicians provided instruction and assistance as needed during this procedure, and participants were ultimately sent home with their device and were informed about scheduled check-in calls to provide support during the monitoring period (see Supplemental Method SM2 for details on instructions during the in-clinic assessment).

Home BP Monitoring

Participants were instructed to measure their BP at home over an 8-day period, taking unsupervised readings twice daily using the programmed HBPM protocol each time, once in the morning between 7:00 AM and 9:00 AM, and once in the evening between 7:00 PM and 9:00 PM, before taking antihypertensive medications. Participants were asked to start within 1 to 2 days of visit 10, but the protocol allowed for a delayed start up to 2 weeks. Participants were asked to sit quietly for 5 minutes before each home measurement, in a calm environment free of distractions. All BP measurements were done on the nondominant arm throughout the protocol, including the 8-days of home monitoring. During the 8-day period, staff conducted 3 check-in calls: (1) reminder to start the home BP assessment, (2) 1 day after starting to confirm initiation and address any concerns, and (3) 6 days after starting as a reminder of the end of the assessment in 2 days and to arrange for device return.

On receipt of the home BP monitors, staff paired each device with the Omron Connect app, allowing stored data to be automatically synced and downloaded. Each downloaded reading was cross-verified against the BP monitor received to ensure data completeness. Following verification, monitors were returned to the participants for personal use. All data were securely uploaded to the Carolina Data Acquisition and Reporting Tool,[24] a centralized platform managed by the study's coordinating center. Initially, data were reviewed monthly, and later on a quarterly basis for quality assurance checks. Field centers received feedback based on these reviews to improve protocol execution.

Quality assurance procedures included manual entry of BP values if syncing with the Omron Connect app failed. All BP readings were cross-referenced with the Omron cloud-based database, using unique device identifiers and in-clinic home device triplicate measurements recorded by staff on the first day of the assessment to accurately match each device to the participant. For analysis, mean SBP and DBP values for each day were averaged across the 8-day period to calculate a representative home BP measurement. Overall, 77% of participants adhered exactly to the protocol, while the remaining participants completed procedures that closely aligned with study expectations.

Primary Outcome: SBP Discordance

We assessed 3 BP discordance outcomes: (1) between AOBP and in-clinic home device (AOBP SBP minus in-clinic home device SBP), (2) between in-clinic home device and HBPM (in-clinic home device SBP minus HBPM SBP), and (3) between AOBP and HBPM (AOBP SBP minus HBPM SBP). This approach was repeated for DBP. The average of the 3 readings for AOBP, the average of 3 readings for the in-clinic home device, and the average of 8-days of readings for HBPM was used for analyses. We defined discordance as a dichotomous variable using an absolute threshold of ±10 mm Hg for SBP. We further categorized SBP discordance directionally as ≥10 mm Hg and ≤−10 mm Hg. This threshold was chosen because it crosses hypertension classification categories and thus could impact clinical management. Additionally, it has a good magnitude corresponding to a moderate to strong effect size.[14] The analysis was repeated for absolute threshold of ±5 mm Hg, ≥5 mm Hg, and ≤−5 mm Hg.

Secondary Outcome: DBP Discordance

DBP discordance was evaluated using absolute thresholds of ±5 mm Hg and ±10 mm Hg. Directional discordance was also assessed as DBP ≥5 mm Hg or ≥10 mm Hg and DBP ≤−5 mm Hg or ≤−10 mm Hg.

Covariates of Interest

Participants self-reported sociodemographic factors, health conditions, and lifestyle activities, and brought medications to clinical exams for staff to record using standardized protocols. Pre-specified covariates of interest included age (in years), sex (male or female), diabetes status, and hypertension medication use. Information on self-reported race (Black or White) was combined with the study center to create a race-center variable categorized as follows: Washington County and White, Jackson and Black, Minneapolis and White, Forsyth and Black, Forsyth and White. Body mass index (BMI) was calculated from measured height (in centimeters) and weight (in kilograms). Arm circumference was determined by measuring the length of the upper arm, recording the midpoint, and measuring the mid-upper arm circumference. We also looked at antihypertensive medication classes and grouped them as first-line and nonfirst-line. This was not adjusted for in our article.

Frail was classified as having 3 or more of 5 validated criteria, pre-frail if they met 1 or 2, and robust if none were met: (1) weight loss, defined as a 10% weight loss since visit 9 or a BMI below 18.5 kg/m^2 at visit 10; (2) exhaustion, determined through self-response in relation to experiencing symptoms some of the time or most of the time with respect to the following Center for Epidemiological Studies-Depression statements: "I felt everything I did was an effort" or "I could not get going"; (3) Slowness, determined as a walking speed in the lowest 20th percentile for sex and height was measured via a 4-meter walk test; (4) weak grip strength, measured as the lowest 20th percentile of grip strength,[25,26] adjusted for sex and BMI; and (5) low energy, a sex-specific composite score calculated from 4 physical activity variables as measured by the Baecke physical activity questionnaire.[27,28] These variables include a summary sports score, leisure sport exercise activity compared with peers, sweat during leisure time, and sports/exercise during leisure activity. In ARIC visit 9, all 4 of these variables were collected. However, in visit 10, only one of the components (the summary sports score) was available. We created a modified low-energy score for visit 10 by using the complete data from visit 9 to establish a generalized linear model that would estimate the low-energy score. More details are provided in Supplemental Methods SM3. A person was categorized as having low energy if their score fell in the lowest 20th percentile.

Statistical Analysis

Cross-tabulations were used to display distributions of SBP and DBP discordance categories across participant subgroups. Participants were characterized into 4 different categories based on SBP: (1) no discordance between AOBP and HBPM, (2) absolute threshold of 10 mm Hg discordance between AOBP and HBPM, (3) >+10 mm Hg discordance between AOBP and HBPM, and (4) <−10 mm Hg difference between AOBP and HBPM. A positive discordance value indicated higher AOBP compared with HBPM, whereas a negative value indicated higher HBPM relative to AOBP. We used Bland-Altman plots to evaluate the agreement between BP measurement modalities[29,30] and included regression lines to evaluate the presence of any proportional bias and LOWESS to characterize trends.

We used logistic regression models to evaluate factors associated with SBP and DBP discordance for each BP comparison (ie, AOBP versus in-clinic home device, in-clinic home device versus HBPM, and AOBP versus HBPM). Covariates were age, BMI (kg/m^2), sex, race-center, diabetes status, hypertension medication use status, arm circumference, and frailty status. Linear regression models were also conducted to evaluate continuous associations between covariates and the magnitude of SBP and DBP discordance with and without adjustment for the covariates above. Agreement between BP modalities was further assessed using Deming regression,[31] Concordance correlation coefficients,[32] and Pearson correlation coefficients. Additional sensitivity analyses included defining SBP discordance using a threshold of 5 mm Hg and restricting protocol assessments to those starting HBPM within 2 days of AOBP. All analyses were conducted using SAS 9.4 and statistical significance was determined at a 2-sided alpha level of 0.05.

RESULTS

Participant Characteristics

A total of 792 participants were included in the analytic sample, with 58% female and a mean age of 84 years. The average BMI was 27.0 kg/m^2, with 42% classified as overweight and 23% as obese. One-third of participants had diabetes, and 83% reported using hypertensive medications. With respect to frailty status, 66% were prefrail, 29% robust, and 5% frail. BP cuff sizes used were 2% small, 78% standard adult, and 20% large (Table 1). In-clinic home device SBP had the highest mean SBP of 134.1 mm Hg, followed by AOBP (130.6 mm Hg) and HBPM (129.6 mm Hg; Table 2). Antihypertensive medication class did not appear to meaningfully influence the likelihood of discordance between AOBP and HBPM measurements (Table S1).

Discordance in SBP

Scatter plots comparing SBP across modalities showed the strongest agreement between AOBP versus in-clinic home device measurements (concordance correlation coefficients=0.678; Pearson r=0.691), with lower concordance observed between AOBP versus HBPM (concordance correlation coefficients=0.536; Pearson r=0.553) and in-clinic home device versus HBPM (concordance correlation coefficients=0.546; Pearson r=0.590; Figure 1). Visual inspection of the scatter plots further suggests that higher home SBP values are associated with greater discordance from AOBP, consistent with a more pronounced white-coat effect at higher BP levels. All comparisons demonstrated proportional bias, with greater differences observed at higher SBP values. Similarly, the Bland-Altman plots revealed variable agreement across SBP modalities, with the smallest bias between AOBP versus HBPM SBP (Figure S2). Larger discrepancies were noted for AOBP versus the in-clinic home device (Figure S3) and the in-clinic home device versus HBPM (Figure S4), particularly at higher SBP values.

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Table 1. Descriptive Characteristics and Absolute SBP Discordance for AOBP Versus HBPM

Categories	N (%; N=792)	No AOBP HBPM discordance (N=405)	±10 mm Hg, AOBP−HBPM discordance (N=387)	+10 mm Hg, AOBP−HBPM discordance* (N=210)	−10 mm Hg AOBP−HBPM discordance* (N=177)
Age, y (mean±SD)	84±3.7				
91–100	47 (6)	23 (49)	24 (51)	10 (21)	14 (30)
86–90	184 (23)	97 (53)	87 (47)	48 (26)	39 (21)
81–85	411 (52)	204 (50)	207 (50)	113 (27)	94 (23)
78–80	150 (19)	81 (54)	69 (46)	39 (26)	30 (20)
Body mass index, kg/m² (mean±SD)	27.0±4.7				
Obesity (≥30)	182 (23)	95 (52)	87 (48)	39 (21)	48 (27)
Overweight (25–<30)	332 (42)	163 (49)	169 (51)	89 (27)	80 (24)
Underweight/normal (<25)	278 (35)	147 (53)	131 (47)	82 (30)	49 (17)
Sex					
Male	333 (42)	170 (51)	163 (49)	72 (22)	91 (27)
Female	459 (58)	235 (51)	224 (49)	138 (30)	86 (19)
Race-center					
Forsyth Black	14 (2)	7 (50)	7 (50)	7 (50)	0 (0)
Forsyth White	161 (20)	76 (47)	85 (53)	53 (33)	32 (20)
Jackson Black	163 (21)	83 (51)	80 (49)	37 (23)	43 (26)
Minneapolis White	256 (32)	144 (56)	112 (44)	69 (27)	43 (17)
Washington White	198 (25)	95 (48)	103 (52)	44 (22)	59 (30)
Diabetes					
Yes	264 (33)	135 (51)	129 (49)	60 (23)	69 (26)
No	528 (67)	270 (51)	258 (49)	150 (28)	108 (21)
Frailty					
Frail	36 (5)	15 (42)	21 (58)	8 (22)	13 (36.1)
Prefrail	524 (66)	258 (49)	266 (51)	138 (26)	128 (25)
Robust	232 (29)	132 (57)	100 (43)	64 (28)	36 (15)
HTN Meds use					
Yes	656 (83)	337 (51)	319 (49)	161 (25)	158 (24)
No	136 (17)	68 (50)	68 (50)	49 (36)	19 (14)
Arm circumference					
Small	12 (2)	6 (50)	6 (50)	5 (42)	1 (8)
Adult	619 (78)	321 (52)	298 (48)	173 (28)	125 (20)
Large	161 (20)	78 (48)	83 (52)	32 (20)	51 (32)

AOBP indicates automated office blood pressure; HBPM, home blood pressure monitoring; HTN Meds, hypertension medications; and SBP, systolic blood pressure.

*Addition of last 2 columns (+10 and −10 mm Hg) gives the ±10 mm Hg.

Discordance in SBP measurements among BP comparisons was common. Discordance ±10 mm Hg occurred most often in the in-clinic home device versus HBPM comparison (50.6%). In contrast, for ±5 mm Hg, discordance was most common in the AOBP versus HBPM comparison (76.1%). Positive discordance (≥10 mm Hg) occurred most in the in-clinic home device versus HBPM comparison (33.5%), while negative discordance (≤-10 mm Hg) was most common in the AOBP versus in-clinic home device comparison (30.1%). A similar pattern emerged with ±5 mm Hg, where positive discordance was again highest for the in-clinic home device versus HBPM comparison (48.6%), and negative discordance was most common in the AOBP versus in-clinic home device comparison (41.8%; Table 3).

The mean difference between AOBP and HBPM SBP varied across several participant characteristics. Notably, males had significantly lower AOBP compared with HBPM SBP (−1.9 mm Hg [95% CI, −3.5 to −0.3]), while females had higher AOBP than HBPM SBP (3.0 mm Hg [95% CI, 1.6–4.5]; *P*-interaction <0.001). Compared with those with large arm circumference (−3.2 mm Hg [95% CI, −5.6 to −0.9]), individuals with small (12.8 mm Hg [95% CI, −3.0 to 28.5]; *P*<0.001), and

Table 2. Mean SBP With SD, Median, Lower, and Upper Quartile Values

Variable	Mean	SD	Median	Lower quartile	Upper quartile
AOBP	130.6 mm Hg	18.1 mm Hg	129.8 mm Hg	117.7 mm Hg	142.0 mm Hg
In-clinic home device	134.1 mm Hg	19.1 mm Hg	132.5 mm Hg	121.3 mm Hg	146.0 mm Hg
HBPM	129.6 mm Hg	14.3 mm Hg	129.2 mm Hg	119.9 mm Hg	138.4 mm Hg

In-clinic home device refers to when the home device was used to measure BP in the clinic setting. AOBP indicates automated office blood pressure; HBPM, home blood pressure monitoring; and SBP, systolic blood pressure.

adult arm circumference sizes (1.8 mm Hg [95% CI, 0.6–3.0]; *P*=0.019) showed significantly higher AOBP relative to HBPM. Frail individuals had lower AOBP relative to HBPM compared with robust participants (−5.1 mm Hg [95% CI, −11.0 to 0.9]; *P*=0.016). This was also observed between in-clinic home device versus HBPM (−9.6 mm Hg [95% CI, −14.9 to −4.3). No significant differences were observed across BMI categories, diabetes status, or most age groups. AOBP and HBPM did not differ meaningfully by antihypertensive medication use (0.1 mm Hg [95% CI, −1.0 to 1.3]; *P*=0.068; Figure 2; Table S2).

SBP discordance of ±10 mm Hg was more common among prefrail adults compared with robust individuals across all 3 comparisons, with particularly higher odds of discordance between AOBP versus in-clinic home device measurements (odds ratio [OR], 2.01 [95% CI, 1.43 to 2.84]). Frail adults had markedly higher odds of a large negative discordance of ≤−10 mm Hg when comparing in-clinic home device measurements versus HBPM (OR, 5.77 [95% CI, 2.31 to 14.40]). Discordance was also more likely among males (OR, 1.47 [95% CI, 1.01 to 2.14]), participants with smaller arm circumference (OR, 8.39 [95% CI, 1.33 to 52.98]), and those with higher age increasing age per 1 year (OR, 1.73 [95% CI, 1.08 to 2.78]; Table 4). When analyses were stratified by protocol adherence, comparing participants who strictly followed the measurement protocol with those who started 1 to 2 days following in-clinic assessment and those who did not adhere entirely to the protocol, the magnitude and direction of these associations remained consistent with no statistically significant differences observed between the groups (Tables S3 and S4).

Comparing AOBP versus HBPM, adults aged 91 to 100 years had lower AOBP compared with those aged 78 to 80 years (β=−5.0 mm Hg [95% CI, −10.06 to 0.001]). This was also observed among males (β=−4.7 mm Hg [95% CI, −6.86 to −2.51]) and frail adults (β=−6.6 mm Hg [95% CI, −11.98 to −1.26]). Small cuffs (versus large) were associated with significantly greater discordance, particularly among AOBP versus HBPM and AOBP versus the in-clinic home device, with the small cuff size showing the largest difference (β=14.4 mm Hg, [95% CI, 4.78 to 24.04]). SBP discordance of ≤−5 mm Hg was more common among frail adults compared with robust individuals across all 3 modalities, with higher odds between AOBP versus HBPM (OR, 4.51 [95% CI, 1.25 to 16.29]). Discordance was also observed among prefrail adults (OR, 1.57 [95% CI, 1.02 to 2.41]), adult arm circumference (OR, 1.89 [95% CI, 1.03 to 3.48]), males (OR, 1.55 [95% CI, 1.06 to 2.27]) and hypertension medication use (OR, 1.83 [95% CI, 1.03 to 3.25]; Tables S5 through S8).

BP discordance was significantly associated with baseline AOBP hypertension status. Participants with baseline AOBP SBP ≥140 mm Hg had significantly higher odds of SBP discordance ≥10 mm Hg across all modality comparisons, particularly between AOBP versus HBPM (OR, 8.27 [95% CI, 5.52 to 12.40]). The largest difference was seen between AOBP versus HBPM (16.9 mm Hg [95% CI, 14.86 to 18.91]). Conversely, they had lower odds of SBP discordance ≤−10 mm Hg

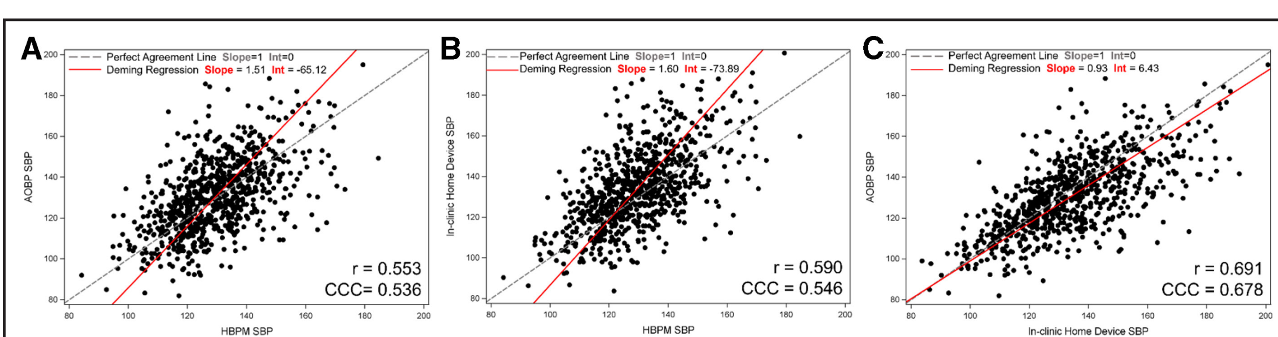

Figure 1. Scatter plot comparing agreement of raw automated office blood pressure (AOBP), home blood pressure monitoring (HBPM), and in-clinic home device measurements using Deming regression, concordance correlation coefficient (CCC), and Pearson correlation (r).

Scatter plots displaying the Deming regression analyses, CCC, and Pearson correlation comparing agreement between systolic blood pressure (SBP) among the following: (**A**) AOBP vs HBPM, (**B**) in-clinic home device vs HBPM, and (**C**) AOBP vs in-clinic home device.

Table 3. SBP Discordance Categories for AOBP Versus HBPM, In-Clinic Home Device Versus HBPM, and AOBP Versus In-Clinic Home Device BP Measurements

	AOBP minus HBPM	In-clinic home device minus HBPM	AOBP minus in-clinic home device
Mean difference in SBP (SD)	1.0 (15.7)	4.5 (15.7)	−3.5 (14.6)
Discordance category	N (%)	N (%)	N (%)
±5 mm Hg discordance	603 (76.1)	596 (75.3)	547 (69.1)
+5 mm Hg discordance	315 (39.8)	385 (48.6)	216 (27.3)
−5 mm Hg discordance	288 (36.4)	211 (26.6)	331 (41.8)
±10 mm Hg discordance	387 (48.9)	401 (50.6)	348 (43.9)
+10 mm Hg discordance	210 (26.5)	265 (33.5)	110 (13.9)
−10 mm Hg discordance	177 (22.4)	136 (17.2)	238 (30.1)

A positive mean difference of AOBP minus HBPM or in-clinic home device minus HBPM means higher AOBP or in-clinic home device compared with HBPM, while a negative mean difference of AOBP minus in-clinic home device means higher in-clinic home device compared with AOBP; in-clinic home device refers to the use of a home device to measure BP in the clinic setting. AOBP indicates automated office blood pressure; BP, blood pressure; HBPM, home blood pressure monitoring; and SBP, systolic blood pressure.

across all modality comparisons (Tables S9 and S10). Across the 8-day HBPM, SBP demonstrated minimal day-to-day variability with an average of 3.27 mm Hg higher morning reading compared with evening (Tables S11 and S12).

Discordance in DBP

DBP across all 3 comparisons also showed the strongest agreement between AOBP versus in-clinic home device measurements with lower concordance observed between HBPM versus AOBP and HBPM versus the in-clinic home device. AOBP versus HBPM and the in-clinic home device versus HBPM demonstrated greater differences at higher DBP values. In contrast, there was a fixed bias observed for AOBP versus the in-clinic home device, with AOBP consistently yielding higher DBP values than the in-clinic home device by a relatively uniform amount across the measurement range (Figure S5). Similarly, Bland-Altman plots revealed variable agreement across DBP modalities with larger discrepancies, particularly at higher DBP values for AOBP versus HBPM and the in-clinic home device versus HBPM, but not AOBP versus the in-clinic home device (Figures S6 through S8). The in-clinic home device had the highest mean of 74.1 mm Hg, followed by HBPM (72.3 mm Hg) and AOBP (64.1 mm Hg; Table S13). Nevertheless, the highest proportions of discordance for both 5 mm Hg (76.5%) and 10 mm Hg (47.5%) were observed with the AOBP versus in-clinic home device comparison (Table S14).

Increasing age was linked to greater DBP discordance between AOBP versus HBPM, as well as AOBP versus in-clinic home device measurement. Adult and small cuffs (versus large) were associated with significantly greater discordance, particularly among AOBP versus HBPM and AOBP versus the in-clinic home device with the small cuff showing the largest difference (β=7.3 mm Hg [95% CI, 2.32–12.25]; Tables S15–S17). Higher BMI was generally associated with greater discordance across comparisons. Use of smaller cuff sizes and frailty status were also associated with discordance with adult cuffs increasing mismatch between AOBP versus the in-clinic home device, and frailty linked to greater differences between the in-clinic home device versus HBPM (Tables S18–S20).

DBP discordance was observed among participants with AOBP DBP ≥90 mm Hg. The most pronounced difference was also observed between AOBP versus HBPM (19.5 mm Hg [95% CI, 13.81 to 25.21]). Similarly, marked DBP discordance ≥5 mm Hg was also observed between AOBP versus HBPM (Tables S9 and S10).

DISCUSSION

In this population of very old adults, discordance between SBP and DBP measurements across BP modalities was common and varied by demographic and clinical characteristics. Although AOBP versus in-clinic home device SBP showed the strongest overall agreement, discordance of ±10 mm Hg was frequently observed, especially between in-clinic home device versus HBPM SBP. Higher resting BP, frailty, smaller cuff size, and male sex were consistently associated with greater discordance. Increasing age was also linked to higher DBP discordance, while higher BMI was associated with reduced discordance. These findings highlight the complexity of interpreting BP measurements among older adults and suggest that physiological changes related to aging and frailty may contribute to inconsistencies across measurements in different environmental settings.

Our study revealed greater discordance between AOBP and HBPM with higher SBP and DBP. Similarly, prior studies have exhibited higher discordance in both SBP and DBP with higher BPs.[33,34] There are several potential explanations. First, the discordance could reflect greater physiological variability related to hypertension among these participants.[35] Moreover, measurement error may be amplified among individuals with a higher resting BP.[36] Alternatively, especially for comparisons between AOBP and HBPM, there is greater precision with HBPM due to averaging more measurements. Taking the difference of 2 variables with unequal precision will result in greater discordance at either tail of the distribution.[33,37] It is also possible that this reflects validation standards that do not emphasize performance in higher BP ranges.[38] In addition, the AOBP protocol used in our study was attended, and prior comparisons of attended versus unattended office BP have shown that attended measurements tend to produce slightly higher BP values, which may have

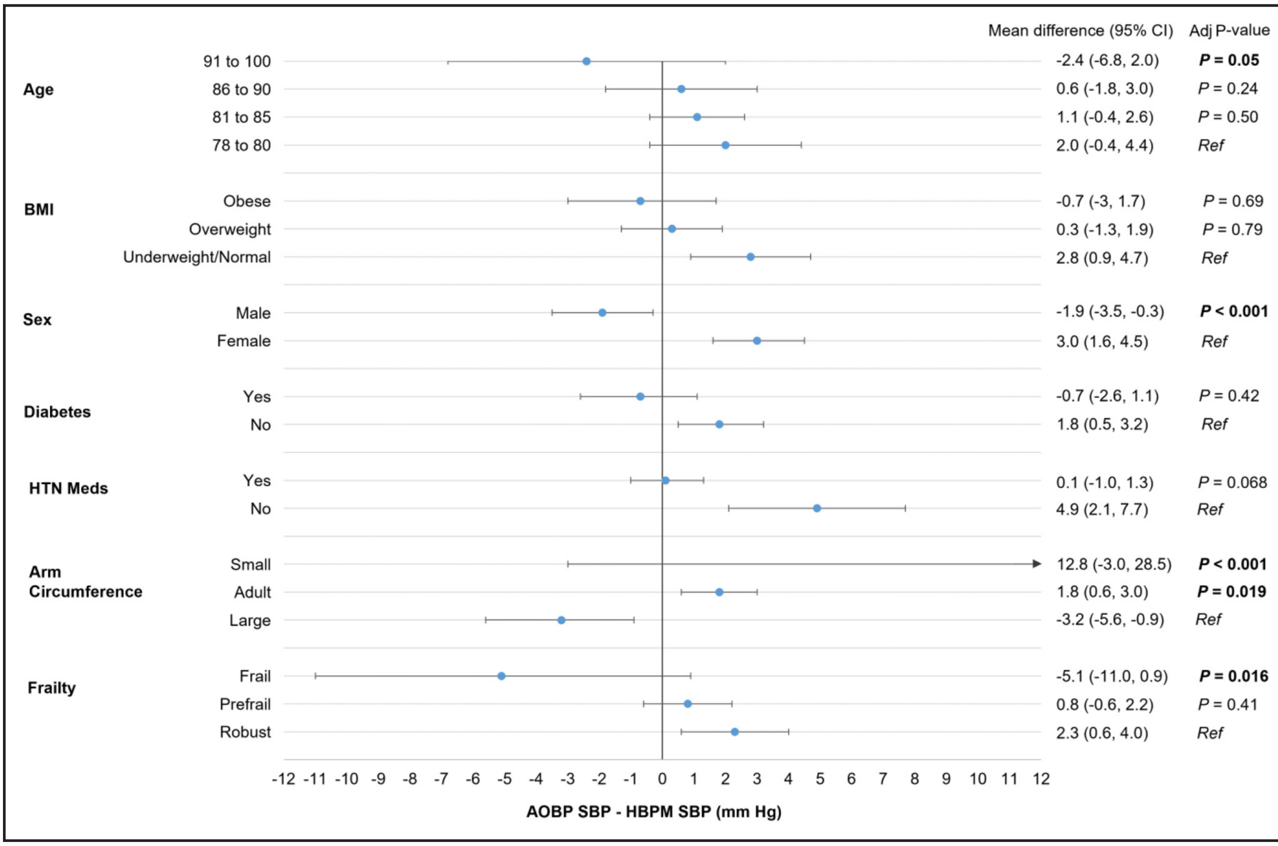

Figure 2. Forest plot of systolic blood pressure (SBP) discordance between automated office blood pressure (AOBP) and home blood pressure monitoring (HBPM) with unadjusted mean differences and adjusted *P* values.

Forest plot showing unadjusted mean difference with 95% CI and adjusted *P* values between AOBP and HBPM stratified by age, body mass index (BMI), sex, diabetes status, hypertension (HTN) medication use, arm circumference, and frailty status. A positive value indicates higher AOBP than HBPM SBP, while a negative value indicates higher HBPM SBP than AOBP. Statistically significant differences ($P<0.05$) are bolded.

contributed to the larger AOBP versus HBPM differences we observed.[39,40] This discordance in the upper range has clinical implications for both the interpretation of hypertension severity and monitoring response to treatment. Whether one modality better identifies patients' cardiovascular disease risk should be the focus of subsequent research.

Frailty status was strongly associated with SBP and DBP discordance across all measurement modalities, with increased frailty associated with higher HBPM values relative to AOBP. Ohata and colleagues also observed frailty to be associated with higher HBPM relative to office BP values in a population of 418 community-dwelling older adults.[41] However, other studies reported no difference in HBPM and office BP among individuals with frailty.[42,43] Differences across studies may reflect the timing of medications relative to home BP measurement.[43] It is also possible that this reflects BP variability, which has been observed to be greater in the setting of frailty.[44] In addition, frail older adults may have more complex pathophysiology, autonomic dysregulation, arterial stiffness, and altered circadian BP regulation.[44–47] Huang et al[48] highlight how biologic mechanisms are strongly associated with frail adults and high resting BP. Another example includes SGLT2 inhibitors, which have been shown to exert favorable effects on BP, physical function, and cardiovascular risk among frail older adults.[49] Although elucidating these mechanisms is beyond the scope of our present study, our findings reaffirm the potential importance of frailty assessments for BP interpretation and cardiovascular risk among older adults.[46]

We found older age, male sex, and smaller cuff size to be associated with greater BP discordance. Associations between age and discordance have been reported by others[15] and may reflect age-related vascular changes and reduced arterial compliance.[50] In contrast, to our knowledge, sex differences have not been reported. We can only speculate as to the cause of this discordance and whether it might be due to differences in technique in the office or at home between male and female participants.[39] Nevertheless, replicating and elucidating these findings should be the focus of subsequent work. Interestingly, cuff size was associated with higher AOBP compared with HBPM measurements. This may be due to the HBPM device in this study having a single size

Table 4. Logistic Regression Comparing SBP Discordance of ±10 mm Hg (OR+95% CI), ≥10 mm Hg (OR+95% CI), and ≤−10 mm Hg (OR+95% CI) between AOBP Versus HBPM, In-Clinic Home Device Versus HBPM, and AOBP Versus In-Clinic Home Device

Variables	AOBP SBP minus HBPM SBP, OR (95% CI)	In-clinic home device SBP minus HBPM SBP, OR (95% CI)	AOBP SBP minus in-clinic home device SBP, OR (95% CI)
SBP discordance of ±10 mm Hg			
91–100 vs 78–80 y	1.21 (0.62–2.38)	1.53 (0.77–3.03)	1.38 (0.68–2.79)
86–90 vs 78–80 y	1.07 (0.68–1.67)	1.11 (0.71–1.75)	1.73 (1.08–2.78)*
81–85 vs 78–80 y	1.26 (0.86–1.85)	1.37 (0.93–2.01)	1.17 (0.78–1.75)
Obese vs underweight/normal	0.87 (0.55–1.37)	0.77 (0.49–1.23)	0.9 (0.56–1.46)
Overweight vs underweight/normal	1.16 (0.83–1.63)	0.9 (0.64–1.26)	0.79 (0.55–1.13)
Male vs female	1.04 (0.77–1.39)	1.04 (0.78–1.40)	0.86 (0.63–1.17)
Diabetes, yes vs no	0.96 (0.70–1.32)	1.02 (0.74–1.40)	1.18 (0.85–1.65)
HTN Meds yes vs no	0.92 (0.62–1.36)	1.13 (0.76–1.68)	1.13 (0.74–1.73)
AC adult vs large	0.79 (0.51–1.22)	0.75 (0.49–1.17)	0.82 (0.52–1.29)
AC small vs large	0.79 (0.22–2.86)	0.84 (0.23–3.14)	2.03 (0.44–9.29)
Frail vs robust	1.90 (0.92–3.93)	2.25 (1.05–4.80)*	0.85 (0.39–1.86)
Prefrail vs robust	1.40 (1.02–1.92)*	1.27 (0.92–1.75)	2.01 (1.43–2.84)*
SBP discordance of ≥10 mm Hg			
91–100 vs 78–80 y	0.88 (0.37–2.11)	1.34 (0.60–2.99)	0.82 (0.27–2.54)
86–90 vs 78–80 y	1.03 (0.60–1.77)	1.10 (0.66–1.84)	1.61 (0.83–3.14)
81–85 vs 78–80 y	1.22 (0.77–1.94)	1.38 (0.89–2.14)	1.00 (0.55–1.83)
Obese vs underweight/normal	0.85 (0.49–1.47)	0.73 (0.43–1.24)	1.41 (0.74–2.71)
Overweight vs underweight/normal	1.16 (0.78–1.74)	0.84 (0.57–1.24)	0.73 (0.43–1.24)
Male vs female	0.74 (0.52–1.06)	0.98 (0.70–1.37)	0.49 (0.30–0.81)*
Diabetes, yes vs no	0.83 (0.56–1.23)	0.87 (0.60–1.25)	1.11 (0.68–1.81)
HTN Meds, yes vs no	0.75 (0.48–1.16)	0.98 (0.63–1.53)	0.98 (0.55–1.76)
AC adult vs large	1.10 (0.63–1.91)	0.73 (0.44–1.22)	2.09 (0.97–4.50)
AC small vs large	1.50 (0.37–6.16)	0.94 (0.22–3.95)	8.39 (1.33–52.98)*
Frail vs robust	1.30 (0.51–3.34)	0.99 (0.37–2.65)	0.96 (0.25–3.63)
Prefrail vs robust	1.21 (0.83–1.77)	1.11 (0.77–1.59)	2.74 (1.56–4.80)*
SBP discordance of ≤−10 mm Hg			
91–100 vs 78–80 y	1.88 (0.81–4.36)	1.95 (0.75–5.04)	1.95 (0.87–4.37)
86–90 vs 78–80 y	1.17 (0.64–2.14)	1.26 (0.64–2.48)	2.21 (1.27–3.87)*
81–85 vs 78–80 y	1.39 (0.83–2.33)	1.40 (0.78–2.53)	1.37 (0.85–2.20)
Obese vs underweight/normal	0.90 (0.50–1.63)	0.87 (0.46–1.67)	0.66 (0.37–1.18)
Overweight vs underweight/normal	1.25 (0.80–1.96)	1.04 (0.64–1.71)	0.74 (0.49–1.13)
Male vs female	1.47 (1.01–2.14)*	1.29 (0.85–1.98)	1.13 (0.79–1.62)
Diabetes, yes vs no	1.14 (0.77–1.68)	1.36 (0.88–2.10)	1.29 (0.88–1.89)
HTN Meds, yes vs no	1.42 (0.80–2.53)	1.48 (0.79–2.77)	1.28 (0.76–2.16)
AC adult vs large	0.64 (0.38–1.09)	0.84 (0.46–1.54)	0.52 (0.30–0.87)*
AC small vs large	0.26 (0.03–2.45)	0.69 (0.07–7.14)	0.74 (0.13–4.14)
Frail vs robust	3.40 (1.43–8.09)*	5.77 (2.31–14.40)*	0.84 (0.34–2.07)
Prefrail vs robust	1.80 (1.16–2.81)*	1.83 (1.10–3.03)*	1.77 (1.19–2.63)*

All models were adjusted for age, BMI, sex, race-center, diabetes status, hypertension medication status, arm circumference, and frailty. In-clinic: using a home device to measure BP in the clinic setting. AC indicates arm circumference; AOBP, automated office blood pressure; BMI, body mass index; HBPM, home blood pressure monitoring; HTN Meds, hypertension medication; OR, odds ratio; and SBP, systolic blood pressure.

*Statistically significant $P<0.05$.

cuff, while the office BP cuff has different cuff sizes. It is well-known that cuff size is a critical feature of accurate BP measurement.[51,52]

Our design allowed us to control some features related to measurement (AOBP versus in-clinic home device) and device (in-clinic home device versus HBPM) settings. Notably, AOBP versus in-clinic home device SBP showed the highest overall agreement, likely due to shared setting and measurement supervision. However, HBPM measurements frequently diverged, particularly underscoring the impact of environmental and behavioral factors, such as measurement conditions, stress, time of day, and unsupervised patient technique.[53] Notably, we observed a fixed bias between AOBP and in-clinic home device DBP. This inaccuracy was also observed when the in-clinic home device measurement was compared with the HBPM average. This suggests that device performance may not explain all the observed discordance between office and HBPM. Moreover, it questions the value of in-clinic verification recommendations, as "device failure" may not be resolved by simply replacing a patient's device.[14,54]

Our study has limitations. First, while the ARIC cohort included a substantial number of very old adults, the sample size for specific subgroups (eg, those classified as frail or using smaller cuff sizes) was limited, which may have reduced statistical power for some comparisons. Second, results may not be generalizable to institutionalized populations or younger adults, as the sample included only community-dwelling older adults. Prior work has shown that discordance may differ with age.[15] Third, the 8-day home BP measurements were unsupervised, which may have introduced some behavioral variability and measurement error despite training. However, these results are likely more similar than different from clinical settings. Fourth, certain physiological confounders, such as stress and medication timing, were not available, limiting our ability to account for temporal variability. Nevertheless, all participants were asked to rest for 5 minutes before measurement and to perform measurements before taking their hypertension medications. However, given that participants may have taken their hypertension medications before their study visit, it is possible that this could be another form of discordance that should be examined further. Fifth, frailty categories used imputation for the low energy criterion to address missing visit 10 data, which may have introduced misclassification. Additionally, frailty components were defined using cohort-specific 20th percentile thresholds for grip strength and slowness, consistent with prior ARIC analyses, which may limit comparability with studies using external normative reference values and could influence estimates of frailty prevalence. Sixth, our study is unable to distinguish positive discordance, that is, a higher office and home measurement, from white coat effects. Last, although we adjusted for multiple demographic and clinical variables, residual confounding remains a possibility.

Our study also has strengths. We used data from a well-characterized, community-based cohort of very old adults, a population often underrepresented in BP measurement research. The inclusion of 3 different BP measurement modalities (AOBP, in-clinic home device, and HBPM) offers a comprehensive evaluation of measurement discordance when controlling for distinct factors (eg, device, environment, user). In addition, all 3 BP modalities were measured among the same individuals, allowing for direct comparisons within participants. High adherence and trained staff support improved the reliability of home BP recordings. BP measurements were obtained using validated devices and standardized protocols modeled after SPRINT, enhancing methodological rigor.[55] Moreover, close coordination with Omron and access to their cloud-based data allowed for thorough adjudication of data. Additionally, ARIC carefully performed a number of assessments relevant to BP and aging, including frailty, BMI, age, and arm circumference, providing clinically relevant insights into the predictors of BP discordance. Finally, the use of multiple concordance and regression methods further strengthened the internal validity and robustness of findings.[56]

The observed variability in BP measurements across settings and subgroups has direct implications for clinical care. Relying on a single modality may misclassify BP control status, particularly in frail or prefrail patients, potentially leading to over- or under-treatment.[6,41] These findings support individualized assessments that incorporate multiple factors such as different BP modalities across different environmental settings, age, frailty/robustness measures, and sex to optimize the precision of hypertension diagnosis, risk assessment, and treatment decisions.[7,10,11] Such an approach is especially critical for very old adults, where the risks of cardiovascular events and hypotension-related complications are heightened.[44,45]

Conclusions

In this population of very old adults, substantial discordance in both SBP and DBP measurements across AOBP, in-clinic home device measurements, and HBPM was common and influenced by resting BP, frailty, age, sex, cuff size, and measurement context. These findings underscore the complexity of accurate BP assessment among older adults and suggest that frailty and physiological aging may play central roles in BP variability. Clinicians should be aware of these discrepancies when making diagnostic or therapeutic decisions based on home BP values among very old patients. Future research should investigate how measurement discordance relates to clinical outcomes and explore whether tailored BP monitoring strategies can improve management among very old and frail adults.

Perspectives

In this cohort of very old adults, we found that nearly half exhibited substantial discordance ≥10 mm Hg between AOBP and HBPM, despite minimal mean differences at the population level. The study highlights the complexity of interpreting BP across different settings in very old adults, where physiological changes, frailty, and cuff size can significantly affect readings. These findings underscore the need to move beyond single-setting measurements and consider individualized, multi-setting BP assessment in older adults. Future research should determine whether discordance predicts cardiovascular or cognitive outcomes and whether measurement-adaptive approaches, such as device-specific calibration or frailty-adjusted thresholds, can improve hypertension management and reduce treatment-related harms in very old adults.

ARTICLE INFORMATION

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Disclosures

None.

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